Neural Networks and Partial Differential Equations

DSA5210 Session 11 —AI for Mathematics

17 June 2026

The curse of dimensionality

Partial differential equations

A **partial differential equation (PDE)** relates an unknown function of several variables to its partial derivatives. Two problems are asked:

- Forward: the equation and its coefficients are known; find u.
- Inverse: some measurements of u are known, but a coefficient, a source term, or the equation itself is not; find the missing piece.

Three running examples for the lecture:

- **Poisson** $-\Delta u = f$ on $[0, 1]$ with $u(0) = u(1) = 0$ — the simplest steady-state PDE; Δ is the Laplacian, the sum of second derivatives.
- **Viscous Burgers** $u_t + u\,u_x = \nu\,u_{xx}$ — the nonlinear term $u\,u_x$ steepens the profile into a near-shock, the viscosity $\nu\,u_{xx}$ smooths it. The standard neural-solver benchmark.
- **Black–Scholes / Hamilton–Jacobi–Bellman (HJB)** in d dimensions — option pricing on d assets, optimal control with d state variables. Genuinely high-dimensional.

Classical solvers and the curse of dimensionality

Classical methods discretize the domain and solve a large algebraic system:

- **finite-difference method (FDM)** — derivatives by difference quotients on a grid;
- **finite-element method (FEM)** — local basis functions on a mesh;
- **spectral methods** — global basis functions, very accurate for smooth solutions.

All place about N points per spatial dimension, hence N^d unknowns in d dimensions. This is the **curse of dimensionality**: a grid of $N = 10$ per axis costs 10^d, which is 10^{100} for $d = 100$. For $d \leq 3$ these methods are fast and accurate; for d in the tens or hundreds they are impossible. High-dimensional PDEs are real — basket Black–Scholes (d assets), HJB (d state variables), the Schrödinger equation for k particles ($3k$ dimensions).

Two settings where a network helps: **high dimension** (no grid, so not automatically N^d) and **inverse problems** with sparse or noisy data. For smooth low-dimensional forward problems, classical methods stay faster and more accurate.

Automatic differentiation

A PDE is a statement about derivatives of u. To check whether a network $u_\theta(x,t)$ satisfies a PDE one needs u_x, u_{xx}, u_t, and so on. **Automatic differentiation** computes them *exactly*: because u_θ is a known composition of elementary operations, the chain rule is applied mechanically through its computational graph.

- Not finite differences (no truncation error) and not symbolic algebra (no expression blow-up).
- The same machinery that trains a network (derivatives with respect to the weights) here also gives derivatives with respect to the *inputs*.

Cheap exact automatic differentiation is the enabling technology. The founding idea is from 1998; the field became active around 2017, when the differentiation libraries (TensorFlow, PyTorch, JAX) matured.

The curse in numbers

For basket Black–Scholes (d assets) or the Schrödinger equation ($3k$ dimensions for k electrons), d runs into the tens or hundreds. A grid with N points per axis has N^d nodes:

dimension d	nodes at $N = 10$	nodes at $N = 100$
1	10	10^2
2	10^2	10^4
3	10^3	10^6
10	10^{10}	10^{20}
100	10^{100}	10^{200}

At $d = 100$, $N = 10$ a grid has 10^{100} nodes — more than the $\approx 10^{80}$ atoms in the observable universe. No finer grid and no faster computer changes this; the obstruction is the exponent d. Monte-Carlo methods replace the grid by M random samples, with error $\propto 1/\sqrt{M}$, *independent of d*.

What makes a PDE hard

Four independent sources of difficulty, each defeating a different classical method:

High dimension the curse above; grids die.

Nonlinearity terms multiply the unknown (Burgers' $u\,u_x$); the algebraic system is nonlinear and solutions can form shocks.

Sharp features shocks, boundary layers, high-frequency oscillation; a uniform grid must be fine everywhere to resolve them.

Irregular geometry a complicated domain is expensive to mesh.

A real problem may combine several. Neural methods target high dimension and mesh-free nonlinearity; sharp features remain hard.

Solving one PDE with a network

Writing the solution as a network

A PDE solution $u(x,t)$ is a function: given any point in the domain, return the field value there.

Classical discretize the domain; solve a linear system for u at N grid points. The solution is N numbers.

Neural represent u as a network $u_\theta(x,t)$: plug in the coordinates, get the field value out. Find θ by training. After training, $u_\theta(x_0,t_0)$ — one forward pass — gives the solution at any point.

Lagaris, Likas, Fotiadis (1998) — the first paper to do this for ODEs and PDEs: write $u(x) = A(x) + F(x)\,N_\theta(x)$, where A satisfies the boundary conditions exactly and F vanishes on the boundary. Train at fixed **collocation points** (locations where the PDE must hold) to minimize the residual. Predated deep learning and modern automatic differentiation; revived in 2019.

Physics-informed neural networks (Raissi, Perdikaris, Karniadakis, 2019)

“Physics-informed”: the known PDE is written into the training loss as a residual penalty — no labeled solution data are needed. Let u_θ be a **multilayer perceptron (MLP)**: inputs (x,t), scalar output u, smooth activation $\tanh$ (so the derivatives the PDE needs exist).

For a PDE $\mathcal{N}[u]=0$ (Burgers: $\mathcal{N}[u]=u_t+u\,u_x-\nu\,u_{xx}$), define the **residual** $r_\theta=\mathcal{N}[u_\theta]$, every derivative by automatic differentiation. The loss is

$$L(\theta)=\underbrace{\tfrac{1}{N_r}\textstyle\sum_i r_\theta(x_i,t_i)^2}_{\text{PDE residual}}+\lambda_b\underbrace{\tfrac{1}{N_b}\textstyle\sum_j(u_\theta-u_{\mathrm{BC}})^2}_{\text{boundary}}+\lambda_0\underbrace{\tfrac{1}{N_0}\textstyle\sum_k(u_\theta-u_{\mathrm{IC}})^2}_{\text{initial}}.$$

The boundary conditions (BC) and initial condition (IC) enter as penalties — **soft constraints**, enforced only approximately, with weights λ_b,λ_0.

PINN: input, output, objective

A physics-informed neural network (PINN):

Input the PDE operator, the domain, the boundary and initial conditions, and collocation points. For a forward problem *no solution data are needed* — the equation supplies the supervision.

Output a trained network u_θ approximating the solution continuously over the whole domain.

Objective minimize the PDE residual plus the boundary/initial mismatch, by Adam (an adaptive-learning-rate gradient method) then L-BFGS (a curvature-using quasi-Newton method for the final sharp convergence).

Benchmark. Burgers with $\nu = 0.01/\pi$, initial condition $u(0, x) = -\sin(\pi x)$ on $[-1, 1]$, zero boundary values. A steep front forms near $x = 0$ at $t \approx 0.4$; the PINN recovers the solution, front included, to a relative L^2 error on the order of 10^{-3} against a high-resolution reference.

The PINN network for Burgers: architecture

The trial function $u_\theta(x,t)$ for the Burgers benchmark is a feed-forward network:

Input $(x,t) \in \mathbb{R}^2$: one space coordinate and one time value.

Hidden layers four layers of 64 neurons each; $\tanh$ activation. $\tanh$ is differentiable to all orders — the PDE residual needs u_x, u_{xx}, and u_t via automatic differentiation.

Output $u_\theta(x,t) \in \mathbb{R}$: the approximate velocity at (x,t).

Parameters $12{,}737$ weights and biases.

A forward pass at any (x,t) gives the approximate solution. The domain $[-1,1] \times [0,1]$ is never discretized into a grid; collocation points are sampled randomly from it during training.

A worked example: a PINN for the Poisson problem

Choose $u^*(x) = \sin(\pi x)$ and derive the forcing term: $f(x) = -u^{*\prime\prime}(x) = \pi^2 \sin(\pi x)$. The PDE to solve is $-u''(x) = \pi^2 \sin(\pi x)$ on $[0, 1]$, $u(0) = u(1) = 0$.

The network $u_\theta(x)$ sees only the **equation and boundary conditions** — not u^*. The residual is $r_\theta(x) = u_\theta''(x) + \pi^2 \sin(\pi x)$ ($\pi^2 \sin(\pi x)$ is the known input f, not the unknown solution). The loss is

$$L(\theta) = \frac{1}{N_r} \sum_i r_\theta(x_i)^2 \; + \; u_\theta(0)^2 \; + \; u_\theta(1)^2.$$

After training, compare u_θ against the known $u^* = \sin(\pi x)$ to measure the error. u^* enters only here, as a yardstick.

A spectral method solves the same problem to machine precision almost instantly; the PINN is slower and less accurate on this 1-D problem.

Soft and hard constraints

The boundary condition can enter the network in two ways.

Soft add it as a penalty term in the loss — the standard PINN. Flexible for any geometry, but it trades off against the residual and is a source of training trouble.

Hard build it into the architecture so it holds exactly. For $u(0) = u(1) = 0$, write $u_\theta(x) = x(1 - x)\,\mathrm{MLP}(x)$, which vanishes at both ends by construction.

Hard constraints remove one loss term and the weight that goes with it, at the cost of a construction tailored to each domain. Soft constraints need no such construction but leave the boundary only approximately satisfied.

Collocation and adaptive sampling

The residual is enforced at sampled interior points — the collocation points.

- Uniform random sampling spends as many points where the solution is smooth as where it is sharp.
- Residual-based adaptive refinement adds points where the residual is currently large — near shocks and boundary layers.

The sharp regions are exactly where the network is least accurate, so concentrating points there is what makes a PINN usable on problems with fronts.

The optimizer and the loss landscape

The loss is non-convex in the weights θ, so there is no guarantee gradient descent reaches the global minimum.

- In practice: Adam first (a robust adaptive-gradient method), then L-BFGS (a curvature-using method) to drive the residual down sharply at the end.
- The loss weights λ_b, λ_0 on the boundary and initial terms matter: set wrong, one term dominates and the others are barely trained.

An approximation theorem can promise a small accurate network exists, yet the optimizer may not find it. The optimizer, not the network's approximation power, is usually what limits a PINN.

PINN convergence: consistency (Shin, Darbon, Karniadakis 2020)

Shin, Darbon, Karniadakis, *Commun. Comput. Phys.* 28 (2020). PDE class: linear second-order elliptic and parabolic with **sign condition** $\tilde{c}(x) = \sum_i D_i b_i(x) + d(x) \leq 0$ (needed for the maximum principle).

Let h_{m_r} minimize the PINN loss within a network class chosen with m_r collocation points, sampled i.i.d. from a fixed distribution on the domain U. As $m_r \to \infty$ (probabilities under the joint law of the samples):

- **Theorem 3.2 — PINN loss rate.** With high probability, $\mathrm{Loss}^{\mathrm{PINN}}(h_{m_r}) = O\left(m_r^{-\alpha/d}\right)$, where $\alpha \in (0, 1)$ is the Hölder exponent of the PDE data and d is the spatial dimension.
- **Theorem 3.3 — consistency.** Almost surely, $h_{m_r} \to u^*$ uniformly on $\overline{U}$ (convergence in C^0). No rate is proved for the C^0 solution error.
- **Theorem 3.4.** If the network class enforces the boundary condition exactly (hard constraint), the convergence strengthens to $H^1(U)$ (functions whose weak gradient is in L^2).

PINN error bound (Mishra, Molinaro 2020)

Mishra, Molinaro, arXiv:2006.16144; *IMA J. Numer. Anal.* 43 (2023).

Theorem 2.6. Let u_θ^* be a PINN trained on N collocation points, $\|\cdot\|_X$ a norm on the solution (e.g. $L^2(D)$ for elliptic, $L^2_t L^2_x$ for parabolic), $p \geq 1$ the residual exponent in the loss. Then

$$\|u - u_\theta^*\|_X \;\leq\; C_{\mathrm{pde}}\, E_T \;+\; C_{\mathrm{pde}}\, C_{\mathrm{quad}}^{1/p}\, N^{-\theta/p},$$

where $E_T = \left(\sum_i w_i |r_{\theta^*}(y_i)|^p\right)^{1/p}$ is the computable training residual; θ the quadrature rate (paper writes α — renamed here to avoid collision with the Hölder α above); for Monte Carlo points $\theta = \frac{1}{2}$, giving $O(N^{-1/(2p)})$; C_{pde} the PDE *stability constant* (how much u changes per unit residual); C_{quad} the quadrature constant.

Small training error $\Rightarrow$ small solution error only when C_{pde} is moderate; an unstable PDE can have a tiny residual and a large solution error.

The Deep Ritz method (E & Yu, 2018)

Many PDEs are the condition for a function to *minimize an energy*. The Poisson problem $-\Delta u = f$ with zero boundary values has the same solution as the minimizer of the **Dirichlet energy**

$$I[u] = \int \left(\tfrac{1}{2} \, |\nabla u|^2 - f \, u \right) dx.$$

Deep Ritz replaces u by a network u_θ, estimates the integral by Monte-Carlo sampling, and minimizes $I[u_\theta]$ over θ. The name is from the **Ritz method** of the calculus of variations — approximating an energy's minimizer within a chosen family, here a deep network.

- Advantage: the energy uses only the first derivative ∇u, not Δu — numerically gentler.
- Cost: a variational (energy) formulation must exist, and the boundary condition is no longer automatic.
- The **Deep Galerkin Method (DGM)** (Sirignano & Spiliopoulos, 2018) is the strong-form sibling — fit the operator and boundary/initial data at randomly sampled points, not a mesh — shown in up to 200 dimensions.

Weak and adversarial formulations (Zang, Bao, Ye, Zhou, 2020)

A **weak adversarial network (WAN)** uses the weak form of the PDE: u solves the PDE if and only if its weak residual against *every* test function φ is zero.

This is rewritten as a min–max (saddle-point) problem:

- the solution is a **primal** network u_θ;
- the worst-case test function is an **adversarial** network φ_η;
- u_θ is trained to minimize, and φ_η to maximize, the weak residual — the structure of a generative adversarial network.

It is mesh-free and aimed at high-dimensional PDEs on irregular domains. The route is distinct: the weak form (one fewer derivative, like Deep Ritz) combined with a *learned* test function in place of a fixed energy.

Backward SDE: the problem

A **forward SDE** gives $X_0 = x_0$ and integrates forward in time — the path is $\mathcal{F}_t$-adapted by construction (it depends only on W_s, $s \le t$).

A **backward SDE (BSDE)** replaces the initial condition with a *terminal* condition and asks for the rest. Find two $\mathcal{F}_t$-adapted processes (Y_t, Z_t) satisfying

$$dY_t = -f(t, X_t, Y_t, Z_t)\,dt + Z_t \cdot dW_t, \qquad Y_T = g(X_T).$$

- $Y_t \in \mathbb{R}$: the value process.
- $Z_t \in \mathbb{R}^d$: the martingale integrand — a second unknown, not auxiliary. By choosing Z_t correctly we steer $\int Z_t\,dW_t$ so that Y_T hits the target $g(X_T)$ without using any future information.

Simplest case $(f = 0)$: $Y_t = \mathbb{E}[g(X_T) \mid \mathcal{F}_t]$ (a martingale); Z_t is given by the martingale representation theorem.

Pardoux–Peng (1990): for f Lipschitz in (y, z) and $g(X_T) \in L^2(\Omega)$, the BSDE has a unique square-integrable adapted solution (Y, Z).

Deep BSDE: the PDE–BSDE equivalence (Han, Jentzen, E, 2018)

This is where a network does what no grid can: solve nonlinear parabolic PDEs in hundreds of dimensions. The mechanism is a classical equivalence.

- A **stochastic differential equation (SDE)** describes a random path X_t driven by Brownian motion.
- A **backward stochastic differential equation (BSDE)** has a terminal condition instead of an initial one; the solution is a pair of adapted processes (Y_t, Z_t) found jointly on $[0, T]$.
- The **nonlinear Feynman–Kac formula**: the solution $u(t, x)$ and its gradient $\nabla u(t, x)$ of a semilinear parabolic PDE are exactly the pair (Y_t, Z_t) solving an associated BSDE along the path X_t.

Deep BSDE: the method

Discretize time $0 = t_0 < \cdots < t_N = T$ and simulate forward paths of X_t (the Euler–Maruyama scheme). Then:

- at each step approximate the gradient $Z_{t_n} = \nabla u(t_n, X_{t_n})$ by a **small network**, one per step — stacked, a single deep network across time;
- the unknown value $u(0, x_0)$ is a trainable scalar;
- run the BSDE forward along each path with the network-supplied gradients; the loss is the squared mismatch between the path's simulated terminal value and the known terminal condition $g(X_T)$.

Input	the PDE (drift, diffusion, nonlinearity, terminal condition) and a starting point x_0.
Output	the value $u(0, x_0)$, and ∇u along sampled paths.
Objective	match the known terminal condition along simulated paths.

Deep BSDE: 100 dimensions

The method was demonstrated on the nonlinear Black–Scholes equation, the Hamilton–Jacobi–Bellman equation, and the Allen–Cahn equation in $d = 100$ dimensions, accurately and at modest cost.

It escapes the curse of dimensionality because it rests on **Monte-Carlo path sampling**: the error is governed by the number of simulated paths and does not grow with the dimension d, unlike a grid whose size is N^d.

From PDE to BSDE: Itô's formula

Target: a semilinear parabolic PDE for $u : \mathbb{R}^d \times [0,T] \to \mathbb{R}$,

$$\frac{\partial u}{\partial t} + \mathcal{L}u + f\Big(t, x, u, \sigma^\top \nabla u\Big) = 0, \quad u(T,x) = g(x),$$

where $\mathcal{L}u = \mu \cdot \nabla u + \frac{1}{2}\,\mathrm{tr}(\sigma\sigma^\top \nabla^2 u)$ is the generator of the SDE $dX_t = \mu\,dt + \sigma\,dW_t$.

Apply Itô's formula to $u(t, X_t)$ along a path of the SDE:

$$d\big[u(t,X_t)\big] = \Big(\tfrac{\partial u}{\partial t} + \mathcal{L}u\Big)dt + (\sigma^\top \nabla u) \cdot dW_t.$$

The PDE says $\frac{\partial u}{\partial t} + \mathcal{L}u = -f(t, X_t, u, \sigma^\top \nabla u)$, so:

$$d\big[u(t,X_t)\big] = -f\Big(t, X_t,\, u(t,X_t),\, \sigma^\top \nabla u(t,X_t)\Big)\,dt + \sigma^\top \nabla u(t,X_t) \cdot dW_t.$$

Set $Y_t := u(t,X_t)$ and $Z_t := \sigma^\top(t,X_t)\nabla u(t,X_t)$. This is exactly a **backward SDE (BSDE)**:

$$dY_t = -f(t, X_t, Y_t, Z_t)\,dt + Z_t \cdot dW_t, \qquad Y_T = g(X_T).$$

Deep BSDE: learning Y_0 and Z_t from paths

Itô's formula gives $Y_0 = u(0, x_0)$ (the PDE solution at the start) and $Z_t = \sigma^\top \nabla u(t, X_t)$ (gradient of u along the path). Neither requires a grid in $\mathbb{R}^d$.

Discretise with N steps, $\Delta t = T/N$, $\Delta W_k \sim \mathcal{N}(0, \Delta t \cdot I_d)$:

$$\begin{aligned} X_{t_{k+1}} &= X_{t_k} + \mu\,\Delta t + \sigma\,\Delta W_k, \\ Y_{t_{k+1}} &= Y_{t_k} - f(t_k, X_{t_k}, Y_{t_k}, Z_{t_k})\,\Delta t + Z_{t_k} \cdot \Delta W_k. \end{aligned}$$

Unknowns:

- $Y_0 \in \mathbb{R}$ — a single scalar (the value to find).
- $Z_{t_k} = \mathcal{Z}_\theta^{(k)}(X_{t_k}) \in \mathbb{R}^d$ — one network per time step.

Loss: $L = \mathbb{E}\big[\,|Y_T - g(X_T)|^2\,\big]$, averaged over simulated paths. After training, $Y_0 \approx u(0, x_0)$.

Two families for high-dimensional PDEs

High-dimensional PDE solvers split into two families (E, Han, Jentzen, review, 2021):

Deep learning represent the solution (or operator) by a network and train it — Deep BSDE, Deep Ritz, Deep Galerkin.

Nonlinear Monte Carlo stochastic recursions with no network — in particular multilevel Picard iteration.

Both sample paths rather than gridding space, so both can avoid the N^d cost. They differ in what carries the approximation: a trained network, or a fixed Monte-Carlo recursion.

Multilevel Picard: a proven curse-free method

Hutzenthaler, Jentzen, Kruse, Nguyen, von Wurstemberger (Proc. R. Soc. A, 2020). For semilinear heat equations with Lipschitz nonlinearity f and Lipschitz terminal condition g: multilevel Picard iteration approximates $u(T,x)$ at any point x in *mean-square* $\left(\mathbb{E}|U_{T,x}^{MLP} - u(T,x)|^2\right)^{1/2} \le \varepsilon$, with computational cost growing only *polynomially* in the dimension d and in $1/\varepsilon$.

A complete proof that these PDEs do not suffer the curse of dimensionality, *with no neural network involved*. It frames the deep-learning methods: the curse can be provably broken. Open: for which PDEs, and whether a network's training actually reaches the small network the existence theorems promise.

The dimension ladder

Three successive algorithmic ideas pushed the empirical record from $d = 100$ to $d = 10^5$, each by fixing a different computational bottleneck. "Theorem?" $=$ end-to-end error bound proven (not just convergence in principle).

d	method	source	theorem?
100	Deep BSDE	Han–Jentzen–E, PNAS 2018	a posteriori only
$10{,}000$	deep splitting	Beck et al., SISC 2021	no
$100{,}000$	SDGD-PINN	Hu et al., NN 2024	no
$1{,}000$	multilevel Picard	Becker et al., CiCP 2020	**yes**
$10{,}000$	multilevel Picard	Neufeld–Wu, SPDE-AC 2025	**yes**

- Rows 1–3: neural-network methods (semilinear parabolic / HJB). Empirical records; the last column shows no end-to-end bound exists yet.
- Rows 4–5: **multilevel Picard** — a Monte Carlo algorithm, not a neural network. Its proofs cover d in the thousands with explicit rates.

$d = 100 \to 10{,}000$: deep splitting

Bottleneck of Deep BSDE at large d: training propagates gradients backward through all N time-step subnetworks simultaneously — one large computational graph. Memory and compute scale with $N \times d$, making $d = 10^4$ infeasible.

Fix (Beck et al., SISC 2021): train one **fresh small network** $V_n(x) \approx u(t_n, x)$ per step, independently, by a single least-squares regression on simulated paths:

$$\text{target:} \quad V_{n-1}(X_{t_{n+1}}) + \Delta t\, f(V_{n-1}, \nabla V_{n-1}).$$

No backpropagation through time; each step is a self-contained regression.

Result: relative error 1.0×10^{-3} at $d = 10{,}000$ on $\partial_t u = \Delta u - \|\nabla u\|^2$, in $1{,}688\,$s on a 2019 gaming GPU (against a Cole–Hopf Monte Carlo reference).

$d = 10{,}000 \to 100{,}000$: stochastic dimension gradient descent

Bottleneck of standard PINN at large d: the PINN residual requires $\Delta u_\theta = \sum_{i=1}^{d} \partial^2 u_\theta / \partial x_i^2$ — that is d second-derivative evaluations per training step, cost $O(d)$, prohibitive at $d = 10^5$.

Fix (Hu et al., Neural Networks 2024): at each step sample a random subset $I \subset \{1, \ldots, d\}$ of coordinates and use

$$\widehat{\Delta} u_\theta = \frac{d}{|I|} \sum_{i \in I} \frac{\partial^2 u_\theta}{\partial x_i^2}$$

as an unbiased estimator of Δu_θ. Cost per step: $O(|I|) \ll O(d)$.

Result: HJB and Schrödinger-type equations in $100{,}000$ effective dimensions in 12 h on a single GPU. No end-to-end error theorem exists for these numbers.

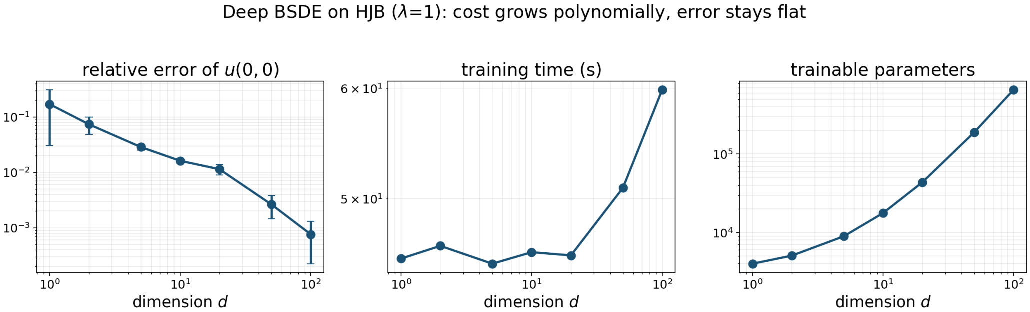
Deep BSDE on the cluster (exp41): HJB in $d = 10$ to 100

Equation (linear-quadratic HJB, Han–Jentzen–E 2018, §4.2): $\partial_t u + \Delta u - \frac{\lambda}{2}\|\nabla u\|^2 = 0$ on $\mathbb{R}^d \times [0,1]$, terminal condition $u(1,x) = g(x) = \ln\left(\frac{1+\|x\|^2}{2}\right)$, $\lambda{=}1$. Quantity of interest: $u(0,\mathbf{0})$ at the origin (reference by 10^7-sample Monte Carlo). Run: $N{=}20$ time steps, $2{,}500$ Adam iterations, 5 seeds, 8 CPU cores.

d	rel. error	wall time
10	1.6%	46 s
50	0.26%	51 s
100	0.076%	60 s

Error stays below 2% across all dimensions tested; wall time grows only from 46 s to 60 s. No grid is built; training cost scales with the number of network parameters, not with the domain size.

The dimension sweep as a picture



Log–log against d: relative error (left), training time (middle), parameter count (right). A grid for $d = 100$ would need 10^{100} nodes; the network trains to 0.076% in one minute on 8 CPU cores. The paper's result: 0.17% in 330 s on a 2017 laptop.

A posteriori error bound from the training loss (Han, Long, 2020)

In $d = 100$ there is usually no exact solution to compare against. A theorem closes this gap for Deep BSDE.

Han–Long (Probability, Uncertainty and Quantitative Risk 5:5, 2020). Let (Y^*, Z^*) solve the BSDE with f Lipschitz in (y, z) and $g(X_T) \in L^2(\Omega)$; let (Y^θ, Z^θ) be the Deep BSDE approximation with N Euler–Maruyama time steps. Then

$$\sup_{t\in[0,T]} \mathbb{E}\big|Y_t^* - Y_t^\theta\big|^2 + \mathbb{E}\int_0^T \big|Z_t^* - Z_t^\theta\big|^2 dt \;\le\; C\Big(\mathbb{E}\big|Y_T^\theta - g(X_T)\big|^2 + \tfrac{1}{N}\Big),$$

with C depending only on T and Lipschitz constants of μ, σ, f.

The right-hand training objective is computable from simulated paths — an *a posteriori* bound. Extension to coupled forward–backward systems with observed non-convergence cases in stochastic control: Negyesi–Huang–Oosterlee, 2024 (arXiv:2403.18552).

A theorem: networks overcome the curse (Grohs, Hornung, Jentzen, von Wurstemberger, 2018)

Statement. Consider linear Kolmogorov PDEs $\partial_t u + \frac{1}{2}\mathrm{tr}(\sigma\sigma^\top \nabla^2 u) + \mu \cdot \nabla u = 0$ with *affine* drift $\mu(x)$ and *affine* diffusion $\sigma(x)$ (Black–Scholes is one case), Lipschitz payoff g at $t = T$, on a bounded box $[a, b]^d$. For every $\varepsilon > 0$ there is a neural network u_θ with

$$\|u(0,\cdot) - u_\theta\|_{L^2([a,b]^d)} \leq \varepsilon, \qquad \#\text{parameters of } u_\theta \ \leq \ C \cdot d^{\,c_1}\, \varepsilon^{-c_2},$$

for absolute constants C, c_1, c_2 (independent of d).

A grid needs ε^{-d} parameters — the curse. The theorem proves a network's representational cost is $\mathrm{poly}(d, 1/\varepsilon)$.

Caveat. Existence only: a small accurate network exists. The theorem does not say gradient descent finds it and does not bound training time. (arXiv 2018; Memoirs of the American Mathematical Society 284(1410), 2023.)

Multilevel Picard on the cluster (exp45, NumPy CPU)

Pure NumPy re-implementation of Hutzenthaler–Jentzen–Kruse–Nguyen–von Wurstemberger (CiCP 2020); no neural network.
Equation class. Semilinear heat $\partial_t u + \Delta u + f(u) = 0$ on $\mathbb{R}^d \times [0, 1)$, terminal $u(1, x) = g(x)$. Allen–Cahn: $f(u) = u - u^3$. Semilinear heat (paper's row): f a sigmoid-type nonlinearity. **QoI**: $u(0, \xi)$ at a fixed reference point ξ (the value below).

equation	d	our value	CiCP 2020
Allen–Cahn	10	0.29589	0.29555
Allen–Cahn	$1{,}000$	0.00339	0.00339
Semilinear heat	$10{,}000$	0.28427	0.28433

Table 1 values reproduced to 3–4 significant figures. Each case runs in seconds on a single CPU core. This is the only solver on the dimension ladder with an end-to-end polynomial-cost proof — and it is not a neural network.

Deep splitting on the cluster (exp46, A100 GPU)

Beck–Becker–Cheridito–Jentzen–Neufeld (SISC 2021).
Equation. HJB $\partial_t u + \Delta u - \|\nabla u\|^2 = 0$ on $\mathbb{R}^d \times [0,1]$, terminal condition $u(1,x) = \varphi(x) = \|x\|^{1/2}$. **QoI**: $u(0,\mathbf{0})$ at the origin; reference by 10^7-sample Monte Carlo.

d	rel. error (ours)	SISC 2021
100	0.20%	0.14%
$1{,}000$	0.30%	0.022%
$10{,}000$	did not converge	0.10%, 1,688 s

At $d = 100$ and $d = 1{,}000$ the method works. The gap at $d = 1{,}000$ reflects hyperparameter sensitivity (the paper uses a longer tuning schedule). At $d = 10{,}000$ we did not reproduce the paper's result; the original run used a 2019 gaming GPU with specific batch-normalisation tuning.

SDGD-PINN on the cluster (exp47, A100 GPU)

Hu–Shukla–Karniadakis–Kawaguchi (Neural Networks 2024).
Equation (Hu et al. §5.1). Sine-Gordon-type semilinear elliptic $\Delta u + \sin(u) = f(x)$ on the unit ball $\{x \in \mathbb{R}^d : \|x\| \leq 1\}$, f chosen so a specified inseparable manufactured solution u^* holds. **QoI**: relative L^2 error against u^* on a held-out set of $20{,}000$ points. SDGD samples $d_{\text{sub}} = 4$ Laplacian terms per step (instead of all d).

d	vanilla PINN	SDGD
100	3.6%	0.63%
$1{,}000$	infeasible	0.059%
$10{,}000$	infeasible	0.24%

Vanilla PINN becomes infeasible above $d = 100$: computing all d Laplacian terms costs $O(d)$ per step. SDGD samples only $d_{\text{sub}} = 4$ terms, staying feasible at any d.

The positive ladder, by strength of statement

1. **Approximation.** Networks of size $\mathrm{poly}(d, 1/\varepsilon)$ *exist* for affine-coefficient Kolmogorov PDEs in $L^2([a,b]^d)$ (Grohs–Hornung–Jentzen–von Wurstemberger, Memoirs of the AMS 284(1410), 2023) and for semilinear heat equations (Hutzenthaler–Jentzen–Kruse–Nguyen, 2020).
2. **Generalization.** Empirical risk minimization needs only $\mathrm{poly}(d, 1/\varepsilon)$ samples for affine-coefficient Kolmogorov equations (Berner–Grohs–Jentzen, SIAM J. Math. Data Sci. 2020).
3. **Conditional variational rates.** Deep Ritz has *dimension-independent* generalization rates in H^1 for Poisson and static Schrödinger problems — provided u^* lies in the **spectral Barron space** $\mathcal{B}^s(\Omega) = \{f : \int (1+|\omega|)^s |\widehat{f}(\omega)|\, d\omega < \infty\}$ (Lu–Lu–Wang, COLT 2021; Barron 1993). The curse is relocated into a regularity assumption, not abolished.
4. **End-to-end algorithmic.** Only multilevel Picard — which is not a neural network.

Missing from every neural rung: a guarantee that the *optimizer* finds the promised network.

The negative results

- **The theory-to-practice gap is a theorem.** For the deep-ReLU approximation classes $\mathcal{N}^{L}_{p,p}$ of fixed depth $L \geq 3$, no algorithm using point samples (SGD included) matches the approximation rate the existence theorems guarantee (Grohs–Voigtlaender, Found. Comput. Math. 24:1085–1143, 2024).
- **Training time can carry the curse.** For shallow ReLU networks trained by gradient flow on a target u^* of Sobolev smoothness r with $2r < d$, the population risk after time t decays no faster than $t^{-4r/(d-2r)}$ — exponentially slow in d (Na–Yang, Information and Inference, 2025).
- **Replication study (McGreivy–Hakim, 2024).** Of ML-for-fluid-PDE papers claiming to beat a standard solver, 79% (60 of 76) compared against a weak baseline; negative replications rarely appear (Nature Machine Intelligence 6:1256–1269). Caveat: that study concerns low-dimensional fluid problems with strong classical baselines; in high dimension no classical baseline exists — the regime's appeal, and why Monte-Carlo cross-checks matter.

The electronic Schrödinger equation

For N electrons and fixed nuclei (Born–Oppenheimer), the Hamiltonian in atomic units is

$$H = -\tfrac{1}{2}\sum_{i=1}^{N}\Delta_{\mathbf{r}_i} - \sum_{i,I}\frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \sum_{i<j}\frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \text{const.}$$

Ground state: $H\psi = E_0\psi$ on $\mathbb{R}^{3N}$ (102 dimensions for bicyclobutane, $N = 34$) with one constraint: ψ must flip sign whenever two electrons are exchanged (Pauli exclusion principle).

Why classical methods fail.

- No grid (10^{100} nodes at $d = 100$, $N = 10$).
- The exact solution **Full CI** expands ψ over all antisymmetric basis functions — $O(N!)$ terms, exponential cost.
- The e–e repulsion $1/|\mathbf{r}_i - \mathbf{r}_j|$ couples all electrons: ψ is not a product of one-electron functions.

Variational Monte Carlo: escaping the exponential

The variational principle: $E_0 \le \langle\psi|H|\psi\rangle/\langle\psi|\psi\rangle$ for any antisymmetric ψ.

The VMC idea. Parametrize a network $\psi_\theta : \mathbb{R}^{3N} \to \mathbb{R}$ that maps all electron coordinates to *one scalar* (the wavefunction value) and *sample* from $|\psi_\theta|^2$:

$$\langle H\rangle_\theta = \mathbb{E}_{\mathbf{R}\sim|\psi_\theta|^2}\left[E_L(\mathbf{R})\right], \quad E_L(\mathbf{R}) = \frac{H\psi_\theta(\mathbf{R})}{\psi_\theta(\mathbf{R})}.$$

Monte Carlo error is $O(1/\sqrt{M})$, independent of d. **No labeled dataset**: the Hamiltonian H is the loss; samples come from the network's own density.

Training loop.

1. Draw $\mathbf{R}_1, \ldots, \mathbf{R}_M \sim |\psi_\theta|^2$ by Metropolis–Hastings MCMC.
2. Compute E_L: kinetic $-\nabla^2\psi/\psi$ via autodiff, potential explicit.
3. Gradient: $\nabla_\theta\langle H\rangle \approx \frac{2}{M}\sum_m (E_L(\mathbf{R}_m) - \langle H\rangle)\nabla_\theta \log\psi_\theta(\mathbf{R}_m)$.
4. Adam step on $W^{(\ell)}, b^{(\ell)}, V_i^k, w_{iI}^k, \pi_{iI}^k$.

The ansatz: from independent electrons to correlated ones

The standard starting point — a single Slater determinant (Hartree–Fock):

$$\psi_{\mathrm{HF}} = \det\big[\phi_i(\mathbf{r}_j)\big]_{i,j=1}^{N}, \quad \phi_i(\mathbf{r}_j) \text{ depends only on } \mathbf{r}_j.$$

Antisymmetry is automatic. But each electron moves in the *average* field of all others — the individual motions are independent. This misses **correlation**: electrons dynamically avoiding each other. The gap $E_{\mathrm{exact}} - E_{\mathrm{HF}}$ is the **correlation energy** — typically $\sim 1\%$ of the total energy, but it governs bond-breaking, dispersion forces, and chemical accuracy.

FermiNet's fix: replace one-electron orbitals with *generalized orbitals* that depend on **all** electrons:

$$\psi_\theta = \sum_{k=1}^{K} \det\big[\phi_i^k(\mathbf{r}_j;\, \{\mathbf{r}_{\neq j}\})\big]_{i,j=1}^{N}.$$

Still antisymmetric (swapping electrons $\leftrightarrow$ swapping determinant columns $\Rightarrow$ sign flip). Now each orbital "knows" where all other electrons are — correlation is captured inside each ϕ_i^k.

FermiNet: how the orbitals see all electrons

Input features. For each electron i: coordinates relative to each nucleus and their distances, spin — forming $\mathbf{h}_i^{(0)}$. For each pair (i,j): relative displacement and distance — $\mathbf{g}_{ij}^{(0)} \in \mathbb{R}^4$.

Four equivariant layers (256 nodes, tanh, residual): at each layer, pool pairwise features spin-selectively and update:

$$\mathbf{h}_i^{(\ell+1)} = \tanh\!\big(W^{(\ell)}[\mathbf{h}_i^{(\ell)};\, \bar{\mathbf{g}}_i^{\uparrow(\ell)};\, \bar{\mathbf{g}}_i^{\downarrow(\ell)}] + b^{(\ell)}\big) + \mathbf{h}_i^{(\ell)},$$

where $\bar{\mathbf{g}}_i^{\uparrow} = \mathrm{mean}_{j:\sigma_j=\uparrow}\mathbf{g}_{ij}$. Permuting electrons permutes the $\mathbf{h}_i$ the same way — equivariant.

Generalized orbital for stream k, spin-orbital i, electron j:

$$\phi_i^k(\mathbf{r}_j) = \underbrace{\left(\sum_I w_{iI}^k e^{-\pi_{iI}^k \|\mathbf{r}_j - \mathbf{R}_I\|}\right)}_{\text{envelope}} \times (V_i^k\, \mathbf{h}_j^{(L)} + c_i^k).$$

$\mathbf{h}_j^{(L)}$ carries information from all electrons: ϕ_i^k captures correlation. The envelope enforces $\psi \to 0$ far from all nuclei.

FermiNet and beyond: accuracy and the timeline

Results (Pfau, Spencer, Matthews, Foulkes, PRR 2:033429, 2020). Bicyclobutane (C_4H_6, 34 electrons, 102 dims): $\sim 97\%$ of the correlation energy — near chemical accuracy. At stretched N_2: more accurate than CCSD(T), quantum chemistry's gold standard method, which fails there.

Key papers since:

- **PauliNet** (Hermann–Schätzle–Noé, Nature Chem. 12:891, 2020): same idea; Hartree–Fock baseline and electron–nuclear cusps built in analytically instead of learned.
- **Psiformer** (von Glehn–Spencer–Pfau, ICLR 2023): self-attention replaces the hand-designed pairwise streams; accuracy gain grows with system size.
- **LapNet** (Li et al., Nature MI 2024): Laplacian in one forward pass, $\sim 10\times$ faster, 116 electrons demonstrated.
- **Orbformer** (arXiv:2506.19960, 2025): pretrained on $22{,}000$ structures, fine-tuned per molecule — the foundation-model pattern applied to the Schrödinger equation.

Learning the solution operator

Solution operators

The **solution operator** of a PDE is the map from its input data to its solution.

- For steady Darcy flow, $G: a(x) \mapsto u(x)$ sends the permeability field to the pressure field.
- For a time-evolution PDE, $G: u(\cdot, 0) \mapsto u(\cdot, T)$ sends the initial state to the state a time T later.

A classical solver recomputes from scratch for every input. Many tasks solve the *same* PDE for thousands of inputs — design optimization, uncertainty quantification, weather — where learning G once and evaluating it cheaply amortizes the cost.

G maps between **function spaces** (infinite-dimensional sets of functions), not between fixed-length vectors. An operator network takes a whole function in and returns a whole function out, independent of the grid each is sampled on.

Universal approximation: from functions to operators

Functions (Cybenko 1989; Hornik 1989; Leshno–Lin–Pinkus 1993). Let σ be a continuous non-polynomial activation, $K \subset \mathbb{R}^n$ compact. For every continuous $f : K \to \mathbb{R}$ and every $\varepsilon > 0$ there exist m, c_i, w_i, b_i with

$$\sup_{x \in K} \Big| f(x) - \textstyle\sum_{i=1}^{m} c_i \, \sigma(w_i \cdot x + b_i) \Big| < \varepsilon.$$

Operators (Chen & Chen 1995, IEEE Trans. Neural Networks 6(4)). Let $K_1 \subset \mathbb{R}^d$, $K_2 \subset \mathbb{R}^{d'}$ be compact, $V \subset C(K_1)$ compact, $G^* : V \to C(K_2)$ a continuous (possibly nonlinear) operator, σ continuous non-polynomial. For every $\varepsilon > 0$ there exist integers m, p, sensor points $x_1, \ldots, x_m \in K_1$, and weights such that

$$\sup_{u \in V,\, y \in K_2} \Big| G^*(u)(y) - \textstyle\sum_{k=1}^{p} \mathrm{branch}_k(u) \cdot \mathrm{trunk}_k(y) \Big| < \varepsilon,$$

$\mathrm{branch}_k(u) = \sum_i c_i^k \sigma\big(\sum_j \xi_{ij}^k u(x_j) + \theta_i^k\big)$ reading u at sensors;
$\mathrm{trunk}_k(y) = \sigma(w_k \cdot y + \zeta_k)$ reading the query point. DeepONet realizes this construction with deep networks.

Training an operator: the data question

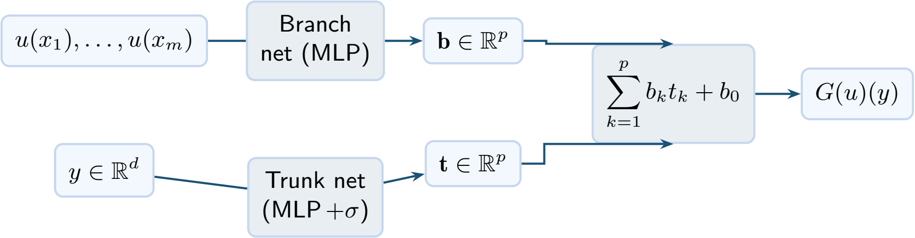
Supervised operator learning needs (input, solution) pairs, and the solutions come from a classical solver run many times in advance.

- The expensive classical solve is paid once, at training time.
- Inference — solving a new instance — is then a single cheap forward pass.

Operator learning fits **many-query** problems (the same PDE for thousands of inputs: optimization, uncertainty quantification, weather) and not a one-off solve, where running the classical solver once would be cheaper than generating a training set.

DeepONet: branch network + trunk network

Two subnets combined by inner product (Lu–Jin–Pang–Zhang–Karniadakis, *Nat. Mach. Intell.* 3:218–229, 2021):



Input	m sensor readings of u (branch) and query point y (trunk).
Output	$G(u)(y) \in \mathbb{R}$: operator value at y.
Training	supervised on (input function, solution) pairs from a classical solver; generalizes to input functions unseen during training.

DeepONet: implementation and named successors

Reference implementation. `DeepXDE` (`github.com/lululxvi/deepxde`; Lu, Meng, Mao, Karniadakis, *SIAM Rev.* 63(1), 2021): PyTorch / TensorFlow / JAX backend, DeepONet and PINN built in. Original DeepONet code at `github.com/lululxvi/deeponet` (Lu, Jin, Pang, Zhang, Karniadakis, *Nature Mach. Intell.* 3, 2021).

Named successors.

- **POD-DeepONet** (Lu et al., 2022): replace the learned trunk with modes from a **proper orthogonal decomposition (POD)** of the training data; often the most accurate variant in the 16-benchmark study below.
- **Physics-informed DeepONet** (Wang, Wang, Perdikaris, 2021): add the PDE residual to the loss; trains with little or no labelled data.
- **Feature-expansion DeepONet** (Lu et al., 2022): add harmonics $(\cos \omega\xi, \sin \omega\xi, \dots)$ to the trunk for oscillatory targets.
- **dFNO+ / gFNO+ / dgFNO+** (Lu et al., 2022) — FNO extensions in the same paper, fitting FNO to different-dimensional inputs/outputs and to complex-geometry domains.

DeepONet error and the operator curse (Lanthaler, Mishra, Karniadakis, 2022)

Setting: separable Hilbert spaces $\mathcal{X}, \mathcal{Y}$ of functions, a probability measure $\mathbb{P}$ on inputs, error measured by $\mathcal{E}(G) = \left(\mathbb{E}_{u\sim\mathbb{P}} \|G(u) - G^*(u)\|_{\mathcal{Y}}^2\right)^{1/2}$. The DeepONet error splits into three pieces (encoder, approximator, reconstructor), each bounded separately.
Two regimes:

- **General Lipschitz operator** $G^* : \mathcal{X} \to \mathcal{Y}$ with $\mathbb{P}$ a centred Gaussian: achieving $\mathcal{E}(G) \leq \varepsilon$ forces a DeepONet of size $\exp(c/\varepsilon^{\beta})$ — the curse of dimensionality reappears at the operator level.
- **Solution operators of specific PDEs** — linear elliptic with Lipschitz coefficient, the heat semigroup, Burgers, Darcy flow, incompressible Navier–Stokes — where the operator inherits the PDE's smoothing/regularity: $\mathcal{E}(G) \leq \varepsilon$ with size only *algebraic* in $1/\varepsilon$.

Neural operators: from kernel integrals to Fourier modes

A **neural operator** maps a whole input function $u(x)$ to a whole output function. Each layer takes a vector-valued function $z(x) : D \to \mathbb{R}^{d_v}$ (the *feature field*, more channels than the original scalar field) and produces another such function.

Kernel integral operator. Given a learned kernel $\kappa : D \times D \to \mathbb{R}^{d_v \times d_v}$, one layer's main step maps

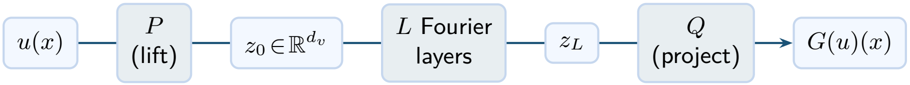
$$z(x) \;\mapsto\; (Kz)(x) \;=\; \int_D \kappa(x,y)\, z(y)\, dy.$$

If $\kappa(x,y) = \kappa(x-y)$ (translation invariant), this is a **convolution** — the same filter is slid across the domain.

Convolution theorem. Convolution in space equals pointwise multiplication in Fourier space: $\mathcal{F}(\kappa * z)(\xi) = \widehat{\kappa}(\xi)\, \widehat{z}(\xi)$. So instead of storing the kernel κ, the **Fourier neural operator (FNO)** stores and learns the kernel's Fourier coefficients $\widehat{\kappa}(\xi)$ directly, one complex weight per kept Fourier mode (Li, Kovachki, Azizzadenesheli, Liu, Bhattacharya, Stuart, Anandkumar, ICLR 2021).

The FNO architecture (Li et al., ICLR 2021)

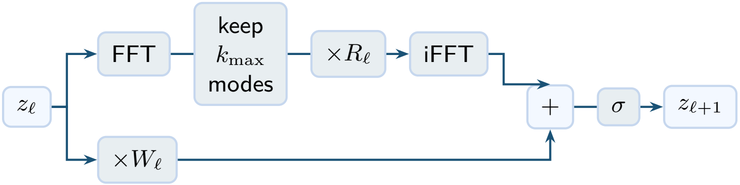
Input: function u on an s-point grid. **Output**: $G(u)$ on the same grid.



- P: **lift** — pointwise MLP, maps scalar $u(x)$ to a d_v-channel feature field $z_0(x)$.
- L **Fourier layers** — each $z_\ell(x) \mapsto z_{\ell+1}(x)$. Mixes information across the domain using frequency modes. See next frame.
- Q: **project** — pointwise MLP, maps d_v channels back to scalar.
- **Resolution invariance**: P, Q act pointwise; Fourier layers act on modes — train at $s = 256$, evaluate at $s = 4096$ with no retraining.

Inside one Fourier layer

Each layer $z_\ell \mapsto z_{\ell+1}$ runs two parallel paths, sums them, then applies activation σ:



FFT / iFFT FFT decomposes the d_v-channel field $z_\ell(x)$ into frequency modes; iFFT transforms back to physical space.

keep $k_{\max}$ modes only the $k_{\max}$ lowest-frequency modes are retained; all higher modes are set to zero.

$\times R_\ell$ each kept mode is multiplied by a learned $d_v \times d_v$ complex matrix. One matrix per mode — these are the **spectral-path trained weights**.

$\times W_\ell$ pointwise linear map in physical space. **Local bypass**: no mixing across space.

Universal approximation for neural operators (Kovachki et al., 2023)

Kovachki, Li, Liu, Azizzadenesheli, Bhattacharya, Stuart, Anandkumar, *JMLR* 24, 2023. Frames the FNO inside a wider class — a **neural operator** is a composition of linear integral operators and pointwise nonlinearities between Banach function spaces — and proves:

Theorem (universal approximation). For every continuous operator $G^* : \mathcal{A} \to \mathcal{U}$ between Banach function spaces on bounded domains, every compact subset $K \subset \mathcal{A}$, and every $\varepsilon > 0$, there exists a neural operator G with

$$\sup_{a \in K} \|G(a) - G^*(a)\|_{\mathcal{U}} \;<\; \varepsilon.$$

Four parameterizations of the integral kernel sit inside this framework: graph kernels, multipole, low-rank, and Fourier (the FNO). The theorem says the framework is expressive enough; it does not bound the network size, nor say that gradient descent finds the small accurate operator the theorem guarantees.

FNO measured: task, data, setup

Task. 1-D viscous Burgers on $[0, 1]$, periodic, viscosity $\nu = 0.1$, terminal time $T = 1$. Operator $G : u_0 \mapsto u(\cdot, 1)$ (initial state to state at $t = 1$).

Data source (`exp42-operator-learning`, course cluster). $1{,}400$ pairs $\big(u_0, u(\cdot, 1)\big)$ generated by an integrating-factor RK4 Fourier spectral solver on $M = 4096$ points, $\Delta t = 10^{-4}$; initial conditions sampled from a **Gaussian random field (GRF)** $u_0 \sim \mathcal{N}(0,\, 625(-\Delta + 25I)^{-2})$ — a random function whose Fourier modes are independent Gaussians with the given covariance. Split: $1{,}200$ train / 200 test.

FNO setup (official `neuraloperator` library, Caltech/NVIDIA): $k_{\max} = 16$ Fourier modes per layer, channel width $d_v = 64$, $L = 4$ layers, $198{,}337$ trainable parameters; 300 epochs (one epoch $=$ one pass over the n_{train} inputs), single random seed; trained at grid resolution $s = 256$.

Metric. Average over the 200 test inputs of the *relative* L^2 error $\big\|G_\theta(u_0) - u(\cdot, 1)\big\|_{L^2} / \|u(\cdot, 1)\|_{L^2}$. *Zero-shot* $=$ evaluated at a grid resolution not used during training, with no retraining.

FNO measured: resolution invariance and the data curve

n_{train}	train time	s=256	s=512	s=1024	s=2048	s=4096
100	29 s	0.0406	0.0404	0.0403	0.0403	0.0402
400	100 s	0.0090	0.0090	0.0090	0.0090	0.0090
1000	253 s	0.0126	0.0123	0.0122	0.0121	0.0121

- Each row constant in s: train at 256, test at 4096 with no retraining and no degradation — resolution invariance, observed directly.
- Data curve: $\sim 4\%$ at $n_{\text{train}} = 100$, $\sim 1\%$ from $n_{\text{train}} = 400$ on (the 1000-input row is slightly worse at this fixed 300-epoch budget, single seed, untuned).
- The `neuraloperator` library (Caltech/NVIDIA) was used; numbers from exp42 on the course cluster.

Amortization measured: 4.21 s per solve vs 3.5 ms per forward pass

- The classical spectral solver (integrating-factor RK4, $M = 4096$, $dt = 10^{-4}$) costs **4.21 s** per new instance — it generated the 1,400-solve training set.
- The trained FNO costs **3.5 ms** per new instance at $s = 4096$ — a factor of $\sim 1{,}200$.
- The one-off cost: 1,400 training solves plus ~ 100 s of training. The saving exists only in the *many-query* regime — amortization, measured.

Python-vs-Python on the same CPU: the order of magnitude is the result, not the digits.

DeepONet measured on the same Burgers operator (exp48)

Same task, data, and split as the FNO row. Vanilla DeepONet via DeepXDE's `DeepONetCartesianProd`: branch / trunk FNN, each 4×128 tanh, $132{,}225$ params; $50{,}000$ Adam ($lr = 10^{-3}$) then $5{,}000$ L-BFGS; single seed; same 16 CPU cores.

n_{train}	train time	rel. L^2 DeepONet	rel. L^2 FNO
100	$1{,}462$ s	0.833	0.0406
400	$1{,}467$ s	0.160	0.0090
$1{,}000$	$1{,}495$ s	0.193	0.0126

Vanilla DeepONet is 15–$20\times$ worse than FNO at every n_{train} on this smooth, periodic, regular-grid task; the $n = 1{,}000$ row is slightly worse than $n = 400$ at this fixed iteration budget. Inference $\sim 33\,\mu$s per instance — branch+trunk dot product, faster than FNO's 3.5 ms at $s = 4096$.

Lu et al. 2022's published DeepONet/FNO parity uses POD-DeepONet with input/output normalization, not vanilla DeepONet.

DeepONet vs FNO: the 16-benchmark study (Lu et al., 2022)

Lu, Meng, Cai, Mao, Goswami, Zhang, Karniadakis, *CMAME* 393:114778, 2022. Same FAIR datasets, same training budget; relative L^2 on the test set. Selection from the 16 benchmarks:

benchmark	DeepONet	POD-DeepONet	FNO
Darcy (rectangle)	1.36%	1.26%	$\mathbf{1.19}\%$
Darcy (pentagram, hole)	1.19%	$\mathbf{0.82}\%$	— (no grid)
Advection II (square wave)	0.27%	$\mathbf{0.08}\%$	54.4%
Navier–Stokes (vort.-vel.)	1.78%	$\mathbf{1.71}\%$	1.81%
Flapping airfoil	$\mathbf{3.65}\%$	—	16.2%

Noise sensitivity (Figs. 6, 7C): add 0.1% Gaussian noise to the input. Advection equation: FNO error jumps to 270%, DeepONet to 0.36%. Instability waves: FNO error rises by $\sim 10{,}000\times$, DeepONet by $\sim 2\times$.

Open benchmark datasets for operator learning

Standardized datasets make methods directly comparable:

PDEBench (Takamoto et al., NeurIPS 2022) Uniform training/test splits and baselines across a broad PDE family (advection, diffusion–reaction, Navier–Stokes, shallow water, …).

PDEArena (Gupta et al., 2022) Over 20 surrogate architectures — FNO, U-Net, Transformers, graph nets — evaluated in a single unified framework.

The Well (Polymathic AI, NeurIPS 2024) 15 TB across 16 datasets from fluid dynamics, plasma physics, and beyond. Designed for pretraining and transfer experiments.

Parametric PDE: the notation $\mathcal{N}_a[u] = 0$

The solution operator $G : a \mapsto u_a$ was introduced in the operators section. Here $a \in \mathcal{A}$ is the parameter that varies from instance to instance (permeability field, initial condition, source term, …).

The compact notation $\mathcal{N}_a[u] = 0$ names the PDE whose solution is u_a. The subscript a means: the differential operator has a as a coefficient or data. Different $a \Rightarrow$ different PDE $\Rightarrow$ different solution.

Two examples.

Darcy flow $\mathcal{N}_a[u] \; = \; -\nabla \cdot (a(x)\, \nabla u) - f$. The subscript a is the permeability field; changing a changes the conductivity and therefore the pressure u.

Burgers equation $\mathcal{N}_a[u] \; = \; u_t + u\, u_x - \nu u_{xx}$ with $u(0, \cdot) = a$. The subscript a is the initial condition; each initial profile gives a different evolution.

PINO trains one FNO $G_\theta \approx G$ on data $\{(a_i, u_i)\}$: trained once, it solves for any a_* in one forward pass.

PINO: training loss and inference (Li et al., arXiv:2111.03794)

Network. An FNO $G_\theta : \mathcal{A} \to \mathcal{U}$.

Training data. N pairs (a_i, u_i) from a classical solver; $N = 0$ is also allowed (physics residual only, no solver data).

Loss.

$$L(\theta) \;=\; \underbrace{\frac{1}{N}\sum_i \|G_\theta(a_i) - u_i\|^2_{L^2}}_{\text{data term}} + \lambda\; \underbrace{\frac{1}{N}\sum_i \|\mathcal{N}_{a_i}[G_\theta(a_i)]\|^2_{L^2}}_{\text{PDE residual}}\,.$$

- The residual $\mathcal{N}_{a_i}[G_\theta(a_i)]$ is computed by automatic differentiation through G_θ.
- The residual can be evaluated on a grid *finer* than the training-data grid: coarse labeled solutions, fine physics.
- With $N = 0$ PINO has only the PDE residual term — and can still converge.

Inference. For a new a_*, one forward pass $G_\theta(a_*)$ returns the solution — same cost as a vanilla FNO query.

FINO: a finite-difference-inspired neural operator (Cheng, Dong, Schönlieb, Aviles-Rivero, 2025)

Position. Hyperbolic PDEs have finite-speed propagation; the solution at (x, t) depends only on a local region. FINO argues that locality should be a hard architectural constraint, not an emergent property of global mixing (FNO, attention). (arXiv:2509.26186)

Local Operator Block (LOB). One layer = three steps:

1. **Learnable stencil** $S(\mathbf{u})[h, w] = \sum_{p,q=-r}^{r} \sum_{c} w_{c,p,q}\, \mathbf{u}_{c,h+p,w+q} + b$ — a $(2r+1)^2$ convolution whose weights replace the fixed finite-difference stencils (e.g. the 5-point Laplacian $[1,-4,1]/h^2$). Same machinery as PDE-Net (Long et al. 2018), now used to *solve* not to *discover*.
2. **Gating mask** $G(\mathbf{u}) = \sigma(W_g * S(\mathbf{u})) \odot S(\mathbf{u})$ — per-location, per-channel importance in $[0, 1]$ suppresses irrelevant derivative responses.
3. **Linear fuse** $\partial_t U_t = W_c * G(\mathbf{u})$.

FINO: explicit time stepping, stability, and results

Explicit time step with learnable Δt: $U_{t+\Delta t} = U_t + \Delta t\,\partial_t U_t$, then $\mathrm{ReLU}(W_p{*}\cdot)$ to form one FINO Block. Three blocks stacked, wrapped in a U-Net.

Composition error bound (Prop. 1). If $\sup_u \|\Psi_\theta(u) - \Phi_{\Delta t}(u)\| \le \varepsilon'$ and $\Phi_{\Delta t}$ is Lipschitz with constant C, then

$$\left\|(\Psi_\theta)^K(u_0) - (\Phi_{\Delta t})^K(u_0)\right\| \;\le\; \tfrac{C^K-1}{C-1}\,\varepsilon' \quad (K\varepsilon' \text{ if } C{=}1).$$

Stability under rollout from per-step approximation quality.
Thm 2 (universal approximation): for every compact $\mathcal{U}$ and $\varepsilon > 0$, a depth-K FD-NET matches $(\Phi_{\Delta t})^K$ uniformly within ε on $\mathcal{U}$.

Results. Up to 44% lower error and $\sim 2\times$ faster than FNO / CNO / transformer baselines on six PDEBench tasks + a climate task.

In-context operator learning: setup (Yang, Liu, Meng, Osher, 2023)

Goal: train *one* network that, at inference, takes a few demos of an unknown PDE/ODE operator and applies it to a new input — no weight update. (arXiv:2304.07993; PNAS 121:11, 2024.)

Vocabulary. *Condition* = the operator's input function (e.g. for the ODE $u' = \alpha u + \beta c + \gamma$: the source $c(t)$ and the initial value $u(0)$). *QoI* (quantity of interest) = the output function $u(t)$. *Demo* = one (condition, QoI) pair. *Prompt* = J demos + one *question condition*. *Query* = keys where the question QoI is asked.

Token representation (key, value, index). Every function is encoded as a sequence of columns; column i holds:

key $\in \mathbb{R}^{d_k}$	term indicator, time coord., space coord., …
value $\in \mathbb{R}$	the function value at that key
index $\in \mathbb{R}^{J+1}$	$+\mathbf{e}_j$ for demo j's condition; $-\mathbf{e}_j$ for demo j's QoI; $\mathbf{e}_{J+1}$ for the question.

Prompt matrix = concatenate all demos' and question's columns along the column axis; column order is irrelevant (the index column tells the model who is who).

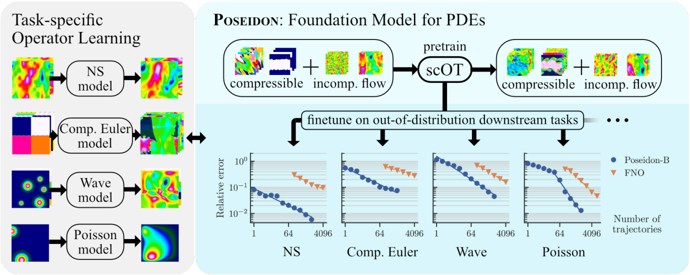
ICON architecture, training, inference

Architecture (DETR-style encoder–decoder; Yang et al. 2023, Fig. 2). Prompt matrix $\to$ shared linear $\to$ **transformer encoder** (self-attention over all prompt columns) $\to$ embedding $\mathbf{E}$ of the unknown operator + question. Queries $\to$ shared linear $\to$ **transformer decoder** (cross-attention into $\mathbf{E}$; *no self-attention* over queries) $\to$ predicted value of the question QoI at each query key.

Training (Algorithm 2). Each step: pick a random problem type + random parameters (defines an operator); sample J (condition, QoI) demos + one question; build prompt, queries, ground-truth values; MSE loss; backprop. J varies across batches (transformer handles variable sequence length).

Inference. New operator $\to$ collect J demos, a question condition, and a list of query keys $\to$ one forward pass returns predictions. The operator is *never* named — it exists only as the embedding $\mathbf{E}$.

Poseidon: a pretrained PDE operator model (Herde et al., NeurIPS 2024)



arXiv:2405.19101; `github.com/camlab-ethz/poseidon`.

What. A neural operator pretrained on a large corpus of fluid-dynamics simulations (compressible and incompressible), then fine-tuned on new PDE types.

Result. On 15 unseen PDE tasks — Navier-Stokes (NS), Compressible Euler, Wave equation, Poisson, ...— Poseidon-B matches an FNO trained from scratch, using *fewer* training trajectories (right panel in figure).

scOT: input, output, the architecture figure

Input. Initial state $a(x) \in C(D; \mathbb{R}^n)$ on $D = [0,1]^2$ (n channels — density, velocity, ...), sampled on a regular grid; *plus* a lead time $t \in [0, T]$. **Output.** Predicted state $S^*_\theta(t, a) \approx u(t, \cdot)$ on the same grid. Sizes: Poseidon-T / -B / -L $= 21$M / 158M / 629M parameters.

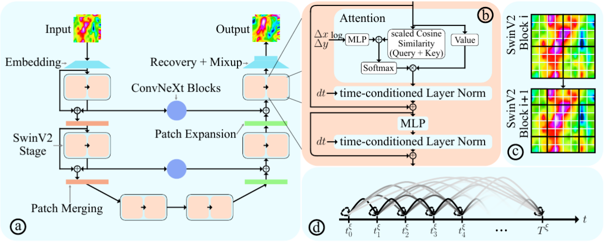


Figure 2: (a) scOT, the model underlying POSEIDON; (b) SwinV2 Transformer block; (c) Shifting Window over patch-based tokens with window (patch) boundaries with black (white); (d) all2all Training for time-dependent PDEs.

Source: Herde et al., arXiv:2405.19101, Figure 2. (a) scOT pipeline; (b) SwinV2 transformer block; (c) shifting window over patch tokens; (d) all2all training.

scOT components (1/2): patch embedding, SwinV2 block, time-LN

Patch embed $\hat{E}$ *Takes* a function $a(x)$ on the grid, n channels. *Does* partition D into $w \times w$ patches; take a weighted average of a inside each patch (one number per channel); pass through a linear layer. *Outputs* a sequence of $(H/w) \times (W/w)$ tokens, each $\mathbf{v} \in \mathbb{R}^C$. *Role*: dense field $\to$ sparse token grid.

SwinV2 block *Takes* a token sequence $\mathbf{v}_{\ell-1}$. *Does* (Eq. 3) $\mathbf{v}'_\ell = \mathbf{v}_{\ell-1} + LN_{\alpha,\beta}(W\text{-MSA}(\mathbf{v}_{\ell-1}))$; $\mathbf{v}_\ell = \mathbf{v}'_\ell + LN_{\alpha,\beta}(\mathrm{MLP}(\mathbf{v}'_\ell))$. **W-MSA** (windowed multi-head self-attention) splits the token grid into $w \times w$ windows; attention runs only inside each window (cost linear in grid size). The window shifts every layer, so over depth every token sees every other one. *Outputs* a token sequence of the same shape. *Role*: cheap spatial mixing.

Time-conditioned LayerNorm *Takes* a feature $\mathbf{v}(x)$ and lead time t. *Does* (Eq. 4) $LN_{\alpha(t),\beta(t)}(\mathbf{v})(x) = \alpha(t) \odot \frac{\mathbf{v}(x) - \mu_{\mathbf{v}}(x)}{\sigma_{\mathbf{v}}(x)} + \beta(t)$ with $\alpha(t) = \alpha t + \bar{\alpha}$, $\beta(t) = \beta t + \bar{\beta}$. *Role*: lead time t enters every block as LN scale/shift; one set of weights handles every $t \in [0, T]$.

scOT components (2/2): hierarchy, skips, recover

Patch merging *Takes* a 2×2 block of neighbouring tokens, each $\in \mathbb{R}^C$. *Does* concatenate into $\mathbb{R}^{4C}$, then linearly project to $\mathbb{R}^{2C}$. *Outputs* 1 token at half spatial resolution and double channels. *Role*: the U-Net down-step — coarser scale gives the next SwinV2 stage a larger effective receptive field for the same window size.

Patch expansion *Takes* 1 token $\in \mathbb{R}^{2C}$. *Does* linearly project to $\mathbb{R}^{4C}$, then reshape into a 2×2 block of tokens $\in \mathbb{R}^C$. *Outputs* 4 tokens at double spatial resolution and half channels. *Role*: the U-Net up-step — restores fine spatial resolution for the output.

ConvNeXt blocks 7×7 depthwise conv $+$ 1×1 channel-mixing MLP (a modern conv block; Liu et al., 2022). Used in *two* places: as the **bottleneck** at the coarsest scale and as the **skip** that connects encoder and decoder features at the same scale.

Recover $\hat{R}$ inverse of $\hat{E}$: linear projection per token, patches un-folded into a continuous output field $u(t, x)$ on the original grid.

Poseidon training: loss, all2all trick, dataset, finetune

Standard loss on M trajectories sampled at $0 = t_0 < \cdots < t_K = T$:

$$\mathcal{L}(\theta) = \frac{1}{M(K+1)} \sum_{i,k} \left\| S^*_\theta(t_k, a_i) - S(t_k, a_i) \right\|^p_{L^p(D)}, \qquad K{+}1 \text{ pairs per trajectory.}$$

all2all loss (Eq. 6). Solution operators satisfy the **semigroup property** $u(t^*) = S(t^* {-} t, u(t))$ for $0 \le t \le t^*$, so every $\bar{k} \le k$ gives a valid pair $(u_i(t_{\bar{k}}), u_i(t_k))$:

$$\widehat{\mathcal{L}}(\theta) = \frac{1}{M\widehat{K}} \sum_{i,\bar{k} \le k} \left\| S^*_\theta(t_k {-} t_{\bar{k}}, u_i(t_{\bar{k}})) - S(t_k {-} t_{\bar{k}}, u_i(t_{\bar{k}})) \right\|^p, \quad \widehat{K} = \tfrac{(K+1)(K+2)}{2}.$$

Quadratic in K: $K{+}1 = 11$ snapshots $\Rightarrow \widehat{K} = 66$ pairs per trajectory.

Pretraining data. 6 operators (4 compressible Euler + 2 incompressible NS), $77{,}840$ trajectories, 11 snapshots each $\Rightarrow \sim 5.11$M pairs.

Finetune. Partition $\theta = [\hat{\theta}, \tilde{\theta}, \tilde{\theta}_{\mathcal{N}}]$: $\hat{\theta}$ (body) initialised from pretraining; $\tilde{\theta}_{\mathcal{N}}$ (time-LN) transferred with higher LR; $\tilde{\theta}$ (embed/recover, when the downstream channel count differs) trained from scratch.

PDEformer-2: input is the PDE itself (Ye et al., Dong group, 2025)

Design point vs Poseidon: Poseidon fixes the PDE family (fluid), varies the initial condition; PDEformer-2 takes the *symbolic PDE itself* as input, so one model handles many PDE forms. (arXiv:2507.15409)

Input: a PDE as a directed acyclic graph (DAG) of nodes and operations, plus coefficient fields, initial condition, and domain shape Ω via SDF_Ω. **Output**: mesh-free $\hat{u}(t, x, y)$ at any query coordinate.

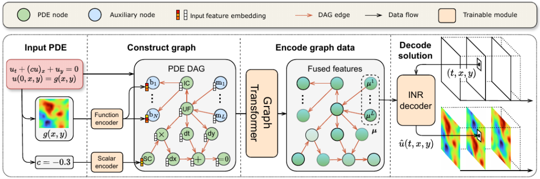


Fig. 1: Overall architecture of PDEformer-2, taking the advection equation $u_t + (cu)_x + u_y = 0$, $u(0, \boldsymbol{r}) = g(\boldsymbol{r})$ on $\Omega = [0, 1]^2$ with periodic boundary conditions as an example. This equation involves only a single unknown field variable $u_1 = u$, and we omit the subscript $j = 1$ for u_j and $\boldsymbol{\mu}_j$ in the figure.

Source: Ye et al., arXiv:2507.15409, Figure 1 (advection example $u_t + (cu)_x + u_y = 0$, $u(0, r) = g(r)$ on $\Omega = [0, 1]^2$ periodic).

PDEformer-2 components: DAG, encoders, Graphormer, Poly-INR

DAG node types `UF` (unknown field), `SC` (scalar coef), `CF` (coefficient field), `VC` (varying coef), `IC` (initial condition), `SDF` (domain shape). **Operation nodes**: `dt`, `dx`, `dy`, `+`, $\times$, $-$, $(\cdot)^2$, $=0$, $\sin$, $\cos$, $\exp_{10}$, $\log_{10}$, `AT` (arbitrary transform).

Scalar encoder 2-layer MLP, 256 hidden units, output dim $d_e = 768$ (the graph embedding dim).

Function encoder CNN on 128×128 field: three stride-4 4×4 convs ($32\to128\to768$ channels), flatten, split into $N{=}4$ branch features.

Graph Transformer (Graphormer; Ying et al. 2021) 12 layers, emb $d_e = 768$, FFN 1536, 32 heads. Initial node embedding $h_i^{(0)} = x_{\mathrm{type}(i)} + \xi_i + z^-_{\deg^-(i)} + z^+_{\deg^+(i)}$. Attention bias $B_{ij} = b^+_{\phi(i,j)} + b^-_{\phi(j,i)} + d_{ij}$ with $\phi(i,j)$ the shortest-path length between nodes; $d_{ij} = -\infty$ for disconnected pairs (masked).

Poly-INR decoder $L = 11$ hidden layers, width 768 (base) or 256 (fast). For each layer ℓ, a hypernet turns μ_j^ℓ into scale/shift modulations $s_\ell^{\mathrm{scale}}, s_\ell^{\mathrm{shift}}$ applied to that layer's input. Activation $\sigma = \sqrt{2}\sin$.

PDEformer-2: pretraining data + use modes

Pretraining dataset ~ 40 TB. Eight *generic* PDE families; each yields many specific PDEs by randomly setting some coefficients to 0 or 1. Data from Dedalus V3 + FEniCSx.

family	samples	data	nRMSE (base)
Diffusion–Convection–Reaction (DCR)	745k	4.85 TB	6.9%
Wave	605k	4.34 TB	6.5%
Multi-variable DCR	660k	12.24 TB	12.6%
Divergence-constrained DCR	400k	7.49 TB	17.0%
MV-Wave / DC-Wave / G-SWE / Elasticity	875k	11.13 TB	3.8–13.0%
Total	3.29**M**	40 **TB**	8.7% avg

Use modes. *Zero-shot*: PDE form within the 8 families, one forward pass returns $\hat{u}$ on the full space-time domain. *Few-shot finetune*: unseen PDE form, < 100 task-specific samples beat specialised neural operators trained from scratch. *Inverse problems*: gradient descent on unknown coefficient scalars or fields with observed u as targets.

Attention patterns in neural PDE solvers

Where the tokens come from decides what attention does:

token	attention scope	cost	example
grid patch	full self-attention	$O(N^2)$	OFormer, GNOT, Transolver
grid patch	$w \times w$ window, shifted per layer	$O(N)$	scOT in Poseidon
DAG node	bias by shortest path $\phi(i,j)$;	$O(\lvert V\rvert^2)$	Graphormer in
	disconnected pairs masked	$\lvert V\rvert \ll N$	PDEformer-2
query coord	cross-attention: query vs encoded input	$O(\lvert q\rvert\lvert \text{in}\rvert)$	HAMLET, ICON

Time conditioning. Same attention weights handle every lead time t by injecting t through scale/shift in the layer norm ($LN_{\alpha(t),\beta(t)}$ in Poseidon's scOT), not into $Q/K/V$ directly.

The opposing camp. FNO replaces global mixing with truncated Fourier modes (no $QK^\top$ at all). FINO (arXiv:2509.26186) argues that for PDEs with finite propagation speed even windowed attention mixes too much; strict locality via learnable finite-difference stencils is the alternative.

Learned weather models

Weather forecasting solves the PDEs of the atmosphere. **Numerical weather prediction (NWP)** integrates them on supercomputers; the operational standard is the Integrated Forecasting System (IFS) of the European Centre for Medium-Range Weather Forecasts (ECMWF). Operator models trained on decades of reanalysis data now match or beat it:

- **FourCastNet** (Pathak et al., 2022) — an adaptive Fourier neural operator at $0.25°$; matches IFS skill at large scales.
- **Pangu-Weather** (Bi et al., *Nature*, 2023) — a 3D transformer; better deterministic 7-day forecasts than the operational IFS, about 10^4 times faster at inference.
- **GraphCast** (Lam et al., *Science*, 2023) — a graph neural network; on par with or better than IFS on nearly all forecast fields.

These are operator surrogates trained on a fixed-resolution Earth, not first-principles solvers, and they inherit their training data's biases.

DeepXDE versus a hand-coded PINN, on identical ground

Burgers' equation $u_t + u\,u_x = \nu u_{xx}$, $\nu = 0.01/\pi$, $u(0,x) = -\sin(\pi x)$ on $[-1,1]$; a $4{\times}64$ $\tanh$ MLP (12,737 parameters), 10,000 collocation points, 15,000 Adam steps followed by L-BFGS (PyTorch 2.x, CPU). DeepXDE 1.15.0 (Lu Lu et al., *SIAM Rev.* 2021) versus the same architecture and optimizer budget, coded from scratch:

implementation	rel. L^2	wall time
hand-rolled (15k Adam + 500 L-BFGS steps)	5.2×10^{-3}	715 s
DeepXDE (15k Adam + L-BFGS to convergence)	3.1×10^{-4}	486 s

The library is $\sim 17\times$ more accurate and $1.5\times$ faster on the same problem. The difference is almost entirely the L-BFGS phase: DeepXDE runs it to its convergence criterion, the hand-rolled version stopped at 500 steps. The optimizer schedule, not the network, was the hand-rolled version's bottleneck.

The high-dimensional toolboxes

- **DeepBSDE** (github.com/frankhan91/DeepBSDE; maintained by Jiequn Han) — the official code, TensorFlow 2, MIT license, still updated; runs the $d = 100$ HJB out of the box from a JSON config.
- **HighDimPDE.jl** (github.com/SciML/HighDimPDE.jl; Julia, MIT, v2.3.0 April 2026) — the only *maintained multi-algorithm* toolkit: Deep BSDE, deep splitting, multilevel Picard, optimal stopping — the neural line and the proven Monte-Carlo line side by side.
- **SDGD_PINN** (github.com/zheyuanhu01/SDGD_PINN; the first author's official PyTorch code) — the 10^5-dimensional scripts.
- **ferminet** (github.com/google-deepmind/ferminet; official, JAX, actively developed) — includes Psiformer, excited states, periodic systems. Molecules also: **deepqmc** (PauliNet lineage); lattice spins: **NetKet**.
- **Mean-field games:** MFGnet.jl (official code of the PNAS 2020 paper) and apac-net — frozen research code.
- No official code exists for deep optimal stopping (Becker–Cheridito–Jentzen); the algorithm is implemented in HighDimPDE.jl.

Spectral bias and training failure

Spectral bias

The approximation theorems say a good network *exists* and is *small*. Training is a separate matter.

An **empirical observation** (Rahaman et al., 2019; later derived in the NTK regime, next frame): networks trained by gradient descent fit the **low-frequency** components of a target first and the **high-frequency** components last, or not at all.

For a PDE whose solution has sharp features — a shock, a boundary layer, a high-frequency oscillation — training plateaus with the coarse shape right and the fine structure wrong. The PINN loss reaches a small value while the error remains concentrated at the shock or boundary layer.

The neural tangent kernel account (Wang, Yu, Perdikaris, 2022)

The **neural tangent kernel (NTK)** makes spectral bias quantitative. In the wide / linearized regime, gradient-descent training behaves like kernel regression with a fixed kernel

$$\Theta(x, x') = \left\langle \nabla_\theta u_\theta(x),\ \nabla_\theta u_\theta(x') \right\rangle.$$

Decompose the error along the eigenvectors of Θ: each component decays at a rate proportional to its **eigenvalue**. The NTK eigenvalues decay rapidly with frequency, so high-frequency error components decay polynomially slowly — spectral bias, derived rather than observed.

A second pathology specific to PINNs: the residual loss and the boundary loss have NTK eigenvalues of very different magnitude, so equal weighting lets one term dominate the gradient and the other is barely trained (the gradient pathologies of Wang & Perdikaris, 2021).

Remedies

- **Loss weighting / learning-rate annealing** (Wang & Perdikaris, 2021): rescale the loss terms during training using their gradient statistics, so no term dominates.
- **Fourier-feature input embeddings** (Tancik et al., 2020; Wang, Wang, Perdikaris, 2021): map the input $x \mapsto [\cos(2\pi Bx), \sin(2\pi Bx)]$ through a bank of frequencies B, so the network sees high frequencies directly and the NTK spectrum flattens.
- Further directions: domain decomposition, causal/curriculum training that respects the time direction, and architectures that build the boundary condition in as a hard constraint.

Reference implementation. `jaxpi` (Wang, Sankaran, Wang, Perdikaris, arXiv:2306.06520, 2023) packages all three remedies — loss re-weighting, Fourier-feature embeddings, and causal training — in a single JAX library with worked examples.

Failure modes beyond frequency (Krishnapriyan et al., 2021)

On convection, reaction, and reaction–diffusion problems, a PINN that solves the easy regime *fails* as a single parameter — for instance the convection speed — grows.

The cause is not limited network capacity. The soft PDE-residual penalty lets the optimizer trade off residual in one region against another; as stiffness grows, spurious local minima proliferate and the global minimum becomes unreachable. Two fixes:

- curriculum training — start from an easy version of the PDE and harden it during training;
- pose the problem as sequence-to-sequence in time, rather than fitting all of space–time at once.

This difficulty is an optimization and loss-landscape problem, distinct from the neural-tangent-kernel spectral bias.

Respecting causality in time (Wang, Sankaran, Perdikaris, 2022)

A time-dependent PINN trained on all times at once can fit late times before the early times are correct — violating the physical arrow of time, and failing on chaotic or turbulent dynamics.

The fix weights the loss so that the residual at a given time is trained only after the earlier times already have small residual — a weight that decays with the accumulated earlier loss.

This restores temporal causality, improves convergence, and provides a quantitative signal of how far the training has progressed in time.

Networks versus finite elements (He, Li, Xu, Zheng, 2020)

A network with the **ReLU** activation (rectified linear unit, $\max(0, x)$) computes a continuous piecewise-linear function — exactly the class that linear finite elements use.

- Every linear-finite-element function in dimension d can be represented by a ReLU network.
- At least two hidden layers are necessary for $d \geq 2$.

So a ReLU network contains the finite-element space and strictly more. The difference is in how the partition is chosen: a network adapts its own piecewise-linear partition through training, where the finite-element method fixes the mesh in advance.

When to use a neural PDE method

Low-dimensional smooth forward problems classical FDM / FEM / spectral methods win on speed and accuracy. Networks do not replace them.

High-dimensional PDEs ($d \gg 3$) classical solvers fail; neural methods (Deep BSDE, deep splitting, SDGD) and multilevel Picard are the available options, with the Grohs et al. theorem behind the proven rung.

Inverse problems, data assimilation, unknown coefficients sparse noisy data; PINNs are natural — physics and data live in one objective.

Many-query settings the same PDE for thousands of inputs; neural operators (FNO, DeepONet) amortize the cost — one training, cheap inference.

Complex geometry networks are mesh-free, avoiding the meshing that strains classical solvers.

Open: reliable training in the presence of spectral bias; error bounds for the network the optimizer actually finds.

What these networks do not do

Neural PDE methods produce numerical solutions and data-driven conjectures. They do not:

- prove existence, uniqueness, or regularity of solutions — the Navier–Stokes global-regularity question, a Millennium Prize problem, is untouched by a network that computes approximate Navier–Stokes flows;
- return a symbolic solution or a theorem — the output is a trained function or a fitted equation, to be verified separately;
- carry an error certificate by default — the approximation theorems say a good network exists, not that the one you trained is good.

Demonstration: three PINN experiments on the cluster

PINNs trained in PyTorch (2.12, CPU) on the course cluster; per-mode error traces and summary numbers recorded for every run.

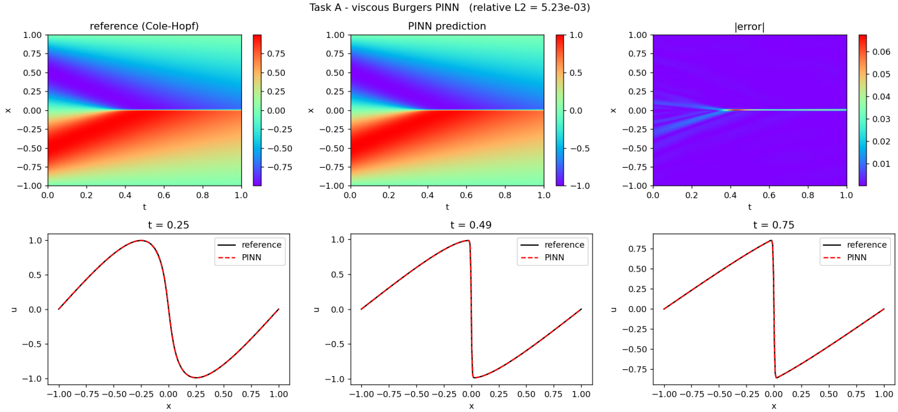
A — Burgers forward Burgers $u_t + u\,u_x = \nu\,u_{xx}$ on $[-1,1]\times[0,1]$, $\nu = 0.01/\pi$, $u(0,x) = -\sin(\pi x)$, $u(t,\pm 1) = 0$; a 4×64 tanh MLP (12,737 params), 10,000 collocation points, 15,000 Adam + 500 L-BFGS steps, ≈12 min on CPU.

B — spectral bias per mode target $u^*(x) = \sin(2\pi x) + \frac{1}{2}\sin(2\pi K x)$; every 400 steps project $u_\theta - u^*$ onto $\sin(2\pi m x)$ and record $|c_m|$. Sweep: {regression, PINN residual} × {plain, Fourier features} × $K \in \{5, 10, 15, 30\}$ × 3 seeds = 48 jobs, 40,000 steps each.

C — inverse recovery $u_t + \lambda_1 u\,u_x = \lambda_2 u_{xx}$ with λ_1, λ_2 trainable; 5,000 noisy samples of the reference solution.

Convergence threshold throughout: $|c_m| \leq 10^{-2}$.

Task A result: Burgers forward solve



Relative L^2 error vs the Cole–Hopf reference: 5.2×10^{-3}; maximum pointwise error 0.068, confined to the thin shock layer at $x = 0$ (top-right panel). The Cole–Hopf reference is more accurate and far cheaper on this 1-D problem.

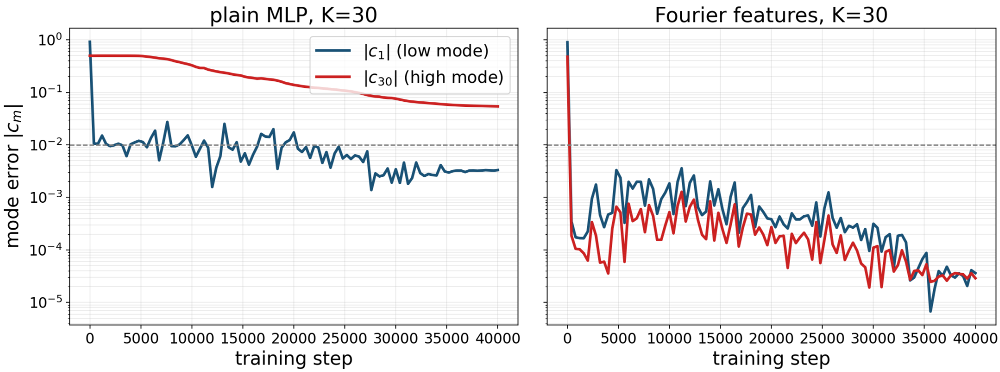
Task B result: steps for each mode to reach 10^{-2} (regression)

Fit u^* by mean-squared error — the pure frequency effect, no operator. Steps for $|c_m| \le 10^{-2}$, mean over 3 seeds:

K	plain, low c_1	plain, high c_K	Fourier, low	Fourier, high
5	400	1,600	400	400
10	1,200	3,600	400	400
15	1,200	7,600	400	400
30	2,000	never (final 5.4×10^{-2})	400	400

The plain network's high mode lags by a factor growing with K — $1{,}600 \to 3{,}600 \to 7{,}600 \to$ never within 40,000 steps — while its low mode converges in a few hundred. Fourier features converge both modes in ~ 400 steps at every K.

Task B: the per-mode error curves ($K = 30$, regression)



Left (plain MLP): $|c_1|$ drops below 10^{-2} within 2,000 steps; $|c_{30}|$ stays near 10^{-1} for all 40,000 steps. Right (Fourier features): both modes fall below 10^{-2} in ~ 400 steps. Dashed line: the 10^{-2} threshold.

Task B, PINN mode: the K^4 conditioning on top of the bias

Train on the residual of $-u'' = f$ instead of on u^*. By Parseval, the L^2 residual is

$$\left\|u_\theta'' + f\right\|_2^2 \;=\; \sum_k (2\pi k)^4 \, |e_k|^2,$$

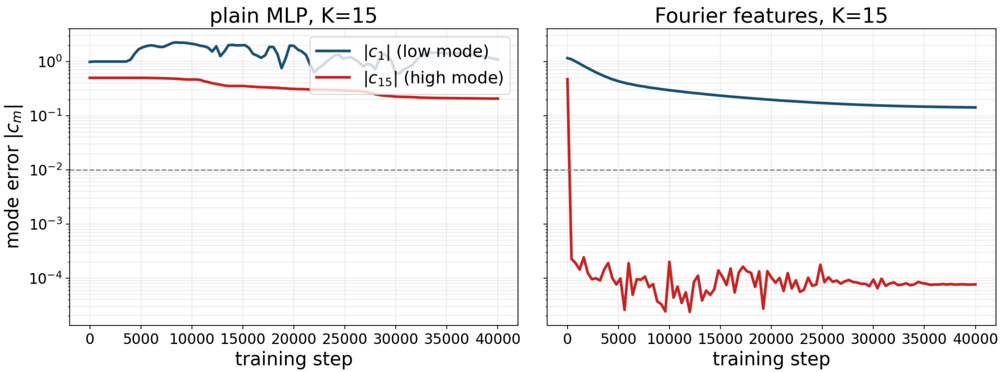
so the mode-k error enters with weight $(2\pi k)^4$ — a loss with condition number $\sim K^4$ on top of the network's frequency bias.

Measured (steps to 10^{-2}, mean over 3 seeds):

- Plain PINN: fails across the family — at $K = 5, 15, 30$ neither mode reaches the threshold in 40,000 steps; at $K = 10$ the high mode crosses at 23,200.
- Fourier features: the high mode converges in ~ 400 steps at every K; at $K \geq 15$ the *low* mode is now the unconverged one (final 0.14 at $K = 15$, 1.06 at $K = 30$).

The imbalance is not removed — it moves to the other end of the spectrum. The full remedy pairs the embedding with loss re-weighting based on the neural tangent kernel.

Task B: the per-mode error curves ($K = 15$, PINN mode)



Left (plain): both modes stuck near 1 — the PINN does not train. Right (Fourier features): $|c_{15}|$ falls below 10^{-4} almost immediately; $|c_1|$ declines slowly and is still ~ 0.14 at 40,000 steps.

Task C result: inverse recovery of (λ_1, λ_2)

$u_t + \lambda_1 u\, u_x = \lambda_2 u_{xx}$; data = 5,000 samples of the reference solution + Gaussian noise; 40,000 Adam + 3,000 L-BFGS steps per noise level.

noise	λ_1 (true 1)	error	λ_2 (true 3.183×10^{-3})	error
0%	0.99717	0.28%	3.468×10^{-3}	8.95%
1%	0.98850	1.15%	4.796×10^{-3}	50.7%
5%	0.99854	0.15%	3.476×10^{-3}	9.21%

- The convection coefficient λ_1 is recovered to $\leq 1.2\%$ at every noise level.
- The viscosity λ_2 — the coefficient of the smallest, highest-curvature term — errs 9–51%, not monotone in the noise (single seed): away from the thin shock layer the data barely distinguish $\nu = 3.2 \times 10^{-3}$ from 4.8×10^{-3}.

Scope of the PINN experiments

- The neural-tangent-kernel spectrum and Fourier-feature eigenvector bias are established results; these experiments measure them on the Burgers equation.
- The Cole–Hopf reference is more accurate and far cheaper than the PINN on these 1-D problems.
- Task C uses synthetic data generated by the reference solver; the governing equation is known in advance.

Discovering the equation from data

PDE-FIND: the method (Rudy, Brunton, Proctor, Kutz, 2017)

Invert the problem: given measurements of $u(x, t)$, find the governing PDE.
PDE-FIND recovers a hidden PDE from a grid of measurements:

1. build a large **library** of candidate terms — u, u_x, u_{xx}, $u\, u_x$, u^2, …— each evaluated from the data (spatial derivatives by finite differences or local polynomial fits);
2. write the time derivative as a linear combination, $u_t \approx \Theta\, \xi$, the columns of Θ the candidate terms, ξ the unknown coefficients;
3. solve by **sparse regression** — STRidge (sequential threshold ridge regression): repeatedly fit and zero out small coefficients — so only a few terms survive.

The surviving terms with their coefficients *are* the discovered PDE; a Pareto trade-off between fit accuracy and the number of active terms selects the model.

PDE-FIND: input, output, results

Input spatiotemporal measurements and a candidate-term library.

Output the few active terms and their coefficients — a PDE.

Objective fit u_t with the fewest terms.

PDE-FIND rediscovers Burgers' equation, the Korteweg–de Vries (KdV) equation, the Schrödinger equation, and Navier–Stokes from data.

It extends **SINDy** (sparse identification of nonlinear dynamics; Brunton, Proctor, Kutz, 2016) from ordinary differential equations to PDEs. Like symbolic regression, it returns a *form* fitted to data, to be verified separately.

Library. `PySINDy` (de Silva et al., *J. Open Source Softw.* 2020; Kaptanoglu et al., 2022) implements SINDy and PDE-FIND in Python; maintained by the Brunton–Kutz group.

Inverse PINNs: coefficients from data

The second route is the inverse mode of the 2019 PINN: write the PDE with its unknown coefficients as **trainable parameters** inside the loss, supply sparse and possibly noisy point observations of u, and train the network and the coefficients together.

- Less general than PDE-FIND — you must guess the *form* of the equation and only fit its coefficients.
- Robust to noise and effective with little data.

Example. Burgers' equation $u_t + \lambda_1 u\, u_x = \lambda_2 u_{xx}$ with unknown (λ_1, λ_2); supply 5,000 noisy pointwise measurements of u; the network and the two coefficients train jointly.

Kolmogorov–Arnold networks (Liu et al., 2024)

PDE-FIND expresses the governing equation as a *sum of library terms* and selects the active ones. KAN offers a complementary route: train a network whose *edges are explicit one-variable functions*, then read the edge shapes as a symbolic formula. Discovery happens by inspecting the trained model directly.

Kolmogorov–Arnold representation theorem (Kolmogorov 1957). For every continuous $f : [0,1]^n \to \mathbb{R}$ there exist $2n+1$ continuous functions $\Phi_q : \mathbb{R} \to \mathbb{R}$ and $n(2n+1)$ continuous monotone functions $\psi_{q,p} : [0,1] \to \mathbb{R}$ such that

$$f(x_1, \ldots, x_n) \ = \ \sum_{q=0}^{2n} \Phi_q\Big(\sum_{p=1}^{n} \psi_{q,p}(x_p)\Big).$$

The inner functions $\psi_{q,p}$ can be chosen *independent of* f (Lorentz 1962; Sprecher 1965).

KAN: architecture and equation-discovery use

An MLP puts fixed activations on the nodes and learns linear weights on the edges. A **KAN** reverses this: nodes sum their inputs; each edge carries a **learnable one-variable spline** $\phi(x)$ instead of a fixed weight.

- A KAN layer: $(\Phi(x))_j = \sum_i \phi_{j,i}(x_i)$, where each $\phi_{j,i}$ is a trainable B-spline (a small set of learnable knot values).
- After training, each edge function $\phi_{j,i}$ can be *plotted or fitted to a known symbolic form* ($\sin$, x^2, e^x, …) and the full network transcribed as a formula.

Evidence (Liu et al., ICLR 2025).

- KANs with $\mathcal{O}(10^2)$ parameters match MLPs with $\mathcal{O}(10^3)$ parameters on PDE solving and physics data-fitting benchmarks.
- KAN trained on known physical laws (special relativity, quantum mechanics) recovers the symbolic expression by visual inspection of edge splines.

Noise-robust discovery, and the network's role

PDE-FIND differentiates the measured data to build its candidate-term library, which amplifies measurement noise. Weak-form variants of sparse regression instead integrate each candidate term against smooth test functions, moving the derivatives onto the known smooth test functions and off the noisy data — far more robust to noise.

Across the discovery methods, the network appears in one of two roles:

- as the regressor — the inverse PINN fitting unknown coefficients;
- as an interpretable function — a Kolmogorov–Arnold network whose edges can be read.

LLM-SR: equation discovery via LLM program proposals (Shojaee et al., ICLR 2025)

Paradigm shift. PDE-FIND fixes a *library* and runs sparse regression. LLM-SR replaces the library by an LLM that proposes whole **equation skeletons as Python programs** $f(\mathbf{x}, \texttt{params})$, then numerically fits the placeholders. (arXiv:2404.18400)

Loop (three steps per iteration):

1. LLM (Mixtral-8x7B or GPT-3.5) gets prompt $=$ problem spec $+$ a few high-scoring past hypotheses; emits b new Python skeletons.
2. Each skeleton runs through `numpy+BFGS` or `torch+Adam` to fit `params`; score $s = -\mathrm{MSE}(\hat{y}, y)$.
3. FunSearch-style island buffer (Romera-Paredes et al., 2024): m populations, Boltzmann sampling, cluster-by-score diversity.

Worked example (nonlinear damped oscillator, "Oscillation 1"): $\dot{v} = F\sin(\omega x) - \alpha v^2 - \beta x^3 - \gamma x v - x\cos(x)$. Mixtral proposes a skeleton with driving / damping / restoring blocks; BFGS fits the constants; best score reached in $\sim 2{,}500$ iterations (vs > 2M for PySR).

LLM-SR vs PySR: out-of-domain error on four benchmarks

Each row is an ODE fitted to training data; both methods are then evaluated on a *held-out input range* not seen during fitting. **OOD NMSE** $= \|\hat{y} - y\|^2 / \|y\|^2$. **Lower is better**; a value > 1 means the formula fails outside the training range.

Task (OOD NMSE ↓)	PySR ($\geq$2M iter)	LLM-SR Mixtral (2.5k iter)
Oscillation 1 — custom nonlinear ODE	0.310	$\mathbf{2{\times}10^{-4}}$
Oscillation 2 — custom nonlinear ODE	$\mathbf{0.010}$	0.029
E. coli bacterial growth ODE	10.14	$\mathbf{0.0037}$
Steel stress-strain (experimental data)	0.130	$\mathbf{0.095}$

Bold = smaller error (better) on that row. LLM-SR wins 3 of 4 tasks with $2{,}500$ iterations vs PySR's $\geq$2M ($800\times$ fewer). On *E. coli*, PySR's extrapolation error (10.14) is $2{,}700\times$ higher than LLM-SR's (0.0037): the evolutionary method found a formula that fits in-distribution but fails outside it. Exception: Oscillation 2, where PySR is $3\times$ more accurate.

References

References: solving one PDE

- I. E. Lagaris, A. Likas, D. I. Fotiadis, *Artificial neural networks for solving ordinary and partial differential equations*, IEEE Trans. Neural Networks 9 (1998) 987–1000.
- M. Raissi, P. Perdikaris, G. E. Karniadakis, *Physics-informed neural networks*, J. Comput. Phys. 378 (2019) 686–707.
- W. E, B. Yu, *The Deep Ritz Method*, Commun. Math. Stat. 6 (2018) 1–12.
- J. Sirignano, K. Spiliopoulos, *DGM: a deep learning algorithm for solving partial differential equations*, J. Comput. Phys. 375 (2018) 1339–1364.
- J. Han, A. Jentzen, W. E, *Solving high-dimensional partial differential equations using deep learning*, PNAS 115 (2018) 8505–8510.
- P. Grohs, F. Hornung, A. Jentzen, P. von Wurstemberger, *A proof that artificial neural networks overcome the curse of dimensionality in the numerical approximation of Black–Scholes PDEs*, Mem. AMS (2023); arXiv:1809.02362.

References: operators, training, discovery

- T. Chen, H. Chen, *Universal approximation to nonlinear operators by neural networks...*, IEEE Trans. Neural Networks 6 (1995) 911–917.
- L. Lu, P. Jin, G. Pang, Z. Zhang, G. E. Karniadakis, *Learning nonlinear operators via DeepONet*, Nat. Mach. Intell. 3 (2021) 218–229.
- Z. Li, N. Kovachki, K. Azizzadenesheli, et al., *Fourier Neural Operator for Parametric PDEs*, ICLR (2021).
- N. Kovachki, Z. Li, et al., *Neural Operator: Learning Maps Between Function Spaces*, JMLR 24 (2023).
- S. Wang, X. Yu, P. Perdikaris, *When and why PINNs fail to train: a neural tangent kernel perspective*, J. Comput. Phys. 449 (2022); and Wang & Perdikaris, SIAM J. Sci. Comput. 43 (2021).
- S. H. Rudy, S. L. Brunton, J. L. Proctor, J. N. Kutz, *Data-driven discovery of partial differential equations*, Sci. Adv. 3 (2017) e1602614.
- G. E. Karniadakis, et al., *Physics-informed machine learning*, Nat. Rev. Phys. 3 (2021) 422–440.

References: theory and recent work

- M. Hutzenthaler, A. Jentzen, T. Kruse, T. A. Nguyen, P. von Wurstemberger, *Overcoming the curse of dimensionality…semilinear parabolic PDEs* (multilevel Picard), Proc. R. Soc. A 476 (2020); W. E, J. Han, A. Jentzen, *…from nonlinear Monte Carlo to machine learning*, Nonlinearity 35 (2021).
- Y. Shin, J. Darbon, G. E. Karniadakis, *On the convergence of PINNs*, Commun. Comput. Phys. 28 (2020); S. Mishra, R. Molinaro, *Estimates on the generalization error of PINNs*, IMA J. Numer. Anal. 43 (2023).
- S. Lanthaler, S. Mishra, G. E. Karniadakis, *Error estimates for DeepONets*, Trans. Math. Appl. 6 (2022).
- A. S. Krishnapriyan, et al., *Characterizing possible failure modes in PINNs*, NeurIPS (2021); S. Wang, S. Sankaran, P. Perdikaris, *Respecting causality…*, arXiv:2203.07404 (2022).
- J. He, L. Li, J. Xu, C. Zheng, *ReLU deep neural networks and linear finite elements*, J. Comput. Math. 38 (2020).
- Y. Zang, G. Bao, X. Ye, H. Zhou, *Weak adversarial networks…*, J. Comput. Phys. 411 (2020); Z. Li, et al., *Physics-Informed Neural Operator*, arXiv:2111.03794 (2021).
- L. Yang, S. Liu, T. Meng, S. J. Osher, *In-context operator learning…*, PNAS 120 (2023); M. Herde, et al., *Poseidon*, NeurIPS (2024); Z. Liu, et al., *KAN: Kolmogorov–Arnold Networks*, ICLR (2025).